Parthiv R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.43
Bhavans Vidyashram Chelembra	Class XII (CBSE)	2021	96.80%
Bhavans Vidyashram Chelembra	Class X (CBSE)	2019	97.40%

EXPERIENCE

Digital Design Intern, Analog Devices, Bengaluru May 2026-Jul 2026

- Studied and analyzed the clocking architecture of a complex multi-clock SoC.
- Performed glitch susceptibility analysis on clock-path multiplexers to verify glitch-free clock switching.
- Analyzed clock distribution networks, clock muxes and clock-domain interactions for robust clock delivery.

Digital Design Intern, Texas Instruments, Bengaluru May 2024-Jul 2024

- Analyzed critical timing paths inside GPTIMER module and proposed timing improvements.
- Developed Python automation scripts for migration from VT FORMAL to RUN FORMAL flow.
- Automated IBIS file generation workflows to improve engineering productivity.

SKILLS

Coding - Verilog, SystemVerilog, Python, C++

Tools & Environment: Cadence Virtuoso, Cadence Simvision, Cadence Xcelium, Xilinx Vivado, Icarus Verilog, LTSpice, Proteus, Linux

Domains: RTL Design, FPGA Design, ASIC Design, Timing Analysis, Computer Architecture, Machine Learning

RELEVANT COURSES

Digital Domain: Digital IC Design, Digital IC Design Lab, Modelling and Testing of Digital Systems, Digital Systems and Circuits, VLSI Circuits and Systems

Computer Architecture: Computer Organization and Architecture, Microprocessors

Analog Domain: Analog IC Design, Analog IC Design Lab

Devices and Fabrication: Semiconductor Device Modelling, VLSI Technology

PROJECTS

M.Tech Thesis Guide: Dr. Shishir Kumar

FPGA-Based Hardware Accelerator for UAV Object Detection June 2026 - Present

- Developing an FPGA-based hardware accelerator for UAV object detection, based on the lightweight SkyNet CNN architecture.
- Developed a software/PyTorch reference model and analyzed the complete inference pipeline from image preprocessing through convolution, feature reorganization, and bounding-box generation.
- Designed and analyzed hardware architectures for depthwise and pointwise convolutions, ReLU6, and max-pooling, including parallelization and resource requirements for FPGA implementation.

High Performance Lane Detection on FPGA June 2024 - April 2025

- Designed an FPGA-based lane detection system using Verilog.
- Implemented grayscale conversion, Gaussian filtering, Sobel edge detection and Hough Transform

16-Stage Pipelined CORDIC Processor March 2026

- Designed a 16-stage pipelined CORDIC processor in Verilog.
- Implemented Rotation and Vectoring modes using fixed-point arithmetic.

Real-Time Power Prediction using Machine Learning January 2026 - May 2026

- Implemented Linear Regression, XGBoost and Random Forest models for real-time power prediction.
- Utilized INA sensor measurements and hardware performance counters on Odroid and Raspberry Pi.

64-Bit Custom SPI Serial-to-Parallel Converter April 2026 - May 2026

- Designed and implemented a 64-bit serial-to-parallel converter using Verilog with a custom SPI-like architecture.
- Verified functionality through simulation and modular implementation using eight 8-bit SPI receiver blocks.

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) -- Probability and Random Variables, Embedded Systems

EXTRACURRICULAR

Sports: Active participation in Football and Badminton